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Singapore Mass Rapid Transit System: planning and implementation

T. W. HULME, BSc, ARSM, FICE, FIStructE, MIMM, MASCE*

P. H. J. CHAPMAN, LLB, BSc, MICE, FIHT, FHKIE, MSIArb, ACIArb*

POK SHEUNG FOO, BE, MIE(Sing), MIE(Aust), MCIT, AMITE†

The three stages of the mass transit studies carried out during the 1970s are described and the principal features of the route selection and its subsequent refinement noted. The establishment of a definitive cost estimate is referred to and the decision to provide the infrastructure as part of the Government public works programme is highlighted. The initial funding proposals and the influence on them of the world-wide downturn in the economy in 1982, when the decision to proceed was taken, are recorded. The roles of government departments and other statutory bodies in the choice and protection of the route are described. The planned implementation programme, together with the economic justification and physical constraints which were involved, is given. Subsequent modifications and the present programme, together with an outline of the further planning of extensions, are also given. The contract arrangements—including contract form, package size, tender procedures, nett present value basis of award and the rationale for acceptance of financing offers—are described in detail and complemented by a review of the administration of the contracts after award. The emphasis given to strict adherence to programme and control of costs, together with the contractors' response, is noted particularly. Progress to date is summarized and some interim conclusions drawn.

Introduction

On the 12 of March 1988, the Prime Minister of Singapore, Mr Lee Kuan Yew, formally opened the Singapore Mass Rapid Transit System. On that day, the full extent of phases 1 and 1A of the system, comprising 20 stations over a 29 km route, became operational and gave Singaporeans their first mass transit railway system. Construction of phase II of the System is well advanced, and staged openings for revenue service began in late 1988.

2. The full extent of the system for which government approval has been given consists of 42 stations over a 67 km route. It will be fully operational by 1990, approximately seven years from the start of construction. The system is shown in Fig. 1.

Early planning

3. In 1967, the Ministry of National Development, with the support of the United Nations Development Programme, commissioned a 4 year state and city

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* Mass Rapid Transit Corporation, Singapore.
† Formerly Mass Rapid Transit Corporation; now Singapore Mass Rapid Transit Ltd.

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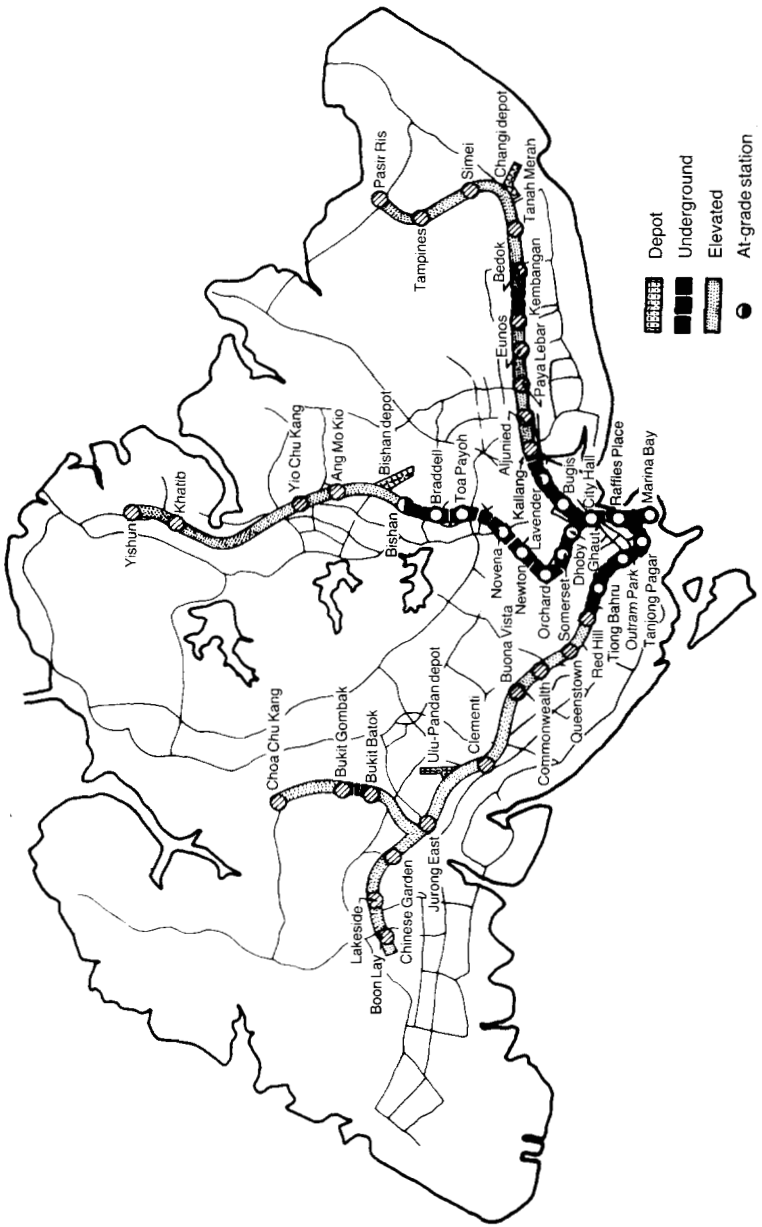


Fig. 1. System under construction

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planning (SCP) project. The purpose of the project was to undertake a comprehensive land-use and transportation study and to formulate recommendations for the long-term physical development of Singapore.

4. Given the limited land resources in Singapore (the main island is only 570 km^2), the relatively large population of approximately 2·6 million, the resulting high density development and the economic profile of the nation, it was clear from the outset that some form of mass transit system would be required. At the same time, it was recognized that the physical limitations would also dictate that the island city state would need to impose restrictions on the growth of car ownership and usage.

5. In the course of the SCP project, advice was sought from London Transport. Based on the transport studies data and proposed land-use development, advisers recommended a conventional rail-based mass rapid transit (MRT) system. These recommendations included: preliminary routing proposals which comprised an east–west line from Siglap (east) to Jurong (west), passing through the central business district; an inner city zone which since 1975 has prevented morning rush-hour entry to cars unless they are carrying four passengers or displaying a paid licence. The line incorporated a branch from Kallang to serve Toa Payoh and other developments in the Thomson Road corridor (Fig. 2).

6. Arising from the SCP project, a Mass Transit Study Unit was established. The unit commissioned three distinct mass transit studies (MTS): MTS phase I, from 1972 to 1974 (funded by the United Nations Development Programme and the Ministry of Communications); MTS phase II, from 1975 to 1976 (funded by the United Nations Development Programme and the Ministry of Communications); MTS phase III, from 1979 to 1980 (funded by the Ministry of National Development).

7. The MTS Unit and the MTS phase I working group comprised a core of local staff, with several members of the SCP project, supplemented by the expertise of several specialist transport consultants (see Table 1). Phase I of the MTS reviewed a number of public transport modes using both road and rail based technologies. Two alternative rail based schemes were proposed. The first was similar to that proposed by the SCP project. The second was a computer generated route based on travel demand forecasts and which, while preserving the SCP east–west line, incorporated a branch to serve the Orchard Road shopping/tourist/commercial corridor before proceeding north to Yio Chu Kang (Fig. 3).

8. The conclusions drawn from the MTS phase I, on which future planning was based, were as follows.

(*a*) A mass rapid transit service in combination with restraints on ownership and use of private cars would provide important economic benefits for Singapore.

(*b*) The rail mode of rapid transit was better suited than other technologies to the anticipated travel demands.

(*c*) The mass rapid transit system should provide direct service between the Orchard Road corridor and the remainder of the central area; therefore, the MTS phase I alternative scheme was to be preferred.

9. The MTS phase II study was commissioned to examine in detail the technical, economic and financial feasibility of the preferred route proposal. The phase II working group was essentially the same as for phase I. Technical studies undertaken by MTS phase II included ground investigation surveys along the proposed

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Table 1. Consultants engaged in planning studies

Study	Period	Consultants
State and city planning study	1967–1971	Crooks, Mitchell, Peacock and Stewart London Transport
Mass transit study phases I and II	1972–1974 1975–1976	Wilbur Smith & Associates Parsons Brinkerhoff Bechtel Tudor
Mass transit study phase III	1979–1980	Mott Hay Anderson & Partners London Transport International Sir William Halcrow & Partners
Harvard review	1980	Harvard University
Comprehensive traffic study	1981	Mott Hay Anderson & Partners Wilbur Smith London Transport International
System design and project staging	1982–1983	Mott Hay Anderson & Partners Sir William Halcrow & Partners London Transport International Wilbur Smith
System design option study	1981–1982	Parsons Brinkerhoff Sofretu
Detailed geotechnical study	1982–1983	Dames & Moore

railway route. As a result of this and other physical restraints which came to light, the proposed route was modified and realigned, particularly in the central area. The most significant modification was to separate the east–west line operationally from the north–south line. Two interchange stations in the central area would facilitate transfers between lines.

10. While no commitment was made by the Government following the MTS phase II study, action was taken by relevant government agencies to safeguard the preferred rail based option. Controls were introduced to prevent development which might later compromise or adversely affect the optimum routeing of the MRT system (infra).

11. The MTS phase III study was initiated some three years after the conclusion of phase II. Phase III, for which a different group of consultants was engaged (Table 1), was required to produce a preliminary engineering design for the preferred MRT system to a level of detail sufficient for the protection of the route, but without imposing unnecessary restrictions on other developments. The track alignments were accurately defined and refinements were incorporated, arising from more precisely defined operational requirements and additional geotechnical information. The north–south line was extended into the newly reclaimed Marina South to provide reversing facilities, and the underground track alignments of the two intersection stations were configured to permit optimum cross–platform interchange (e.g. southbound/eastbound at City Hall; southbound/westbound at

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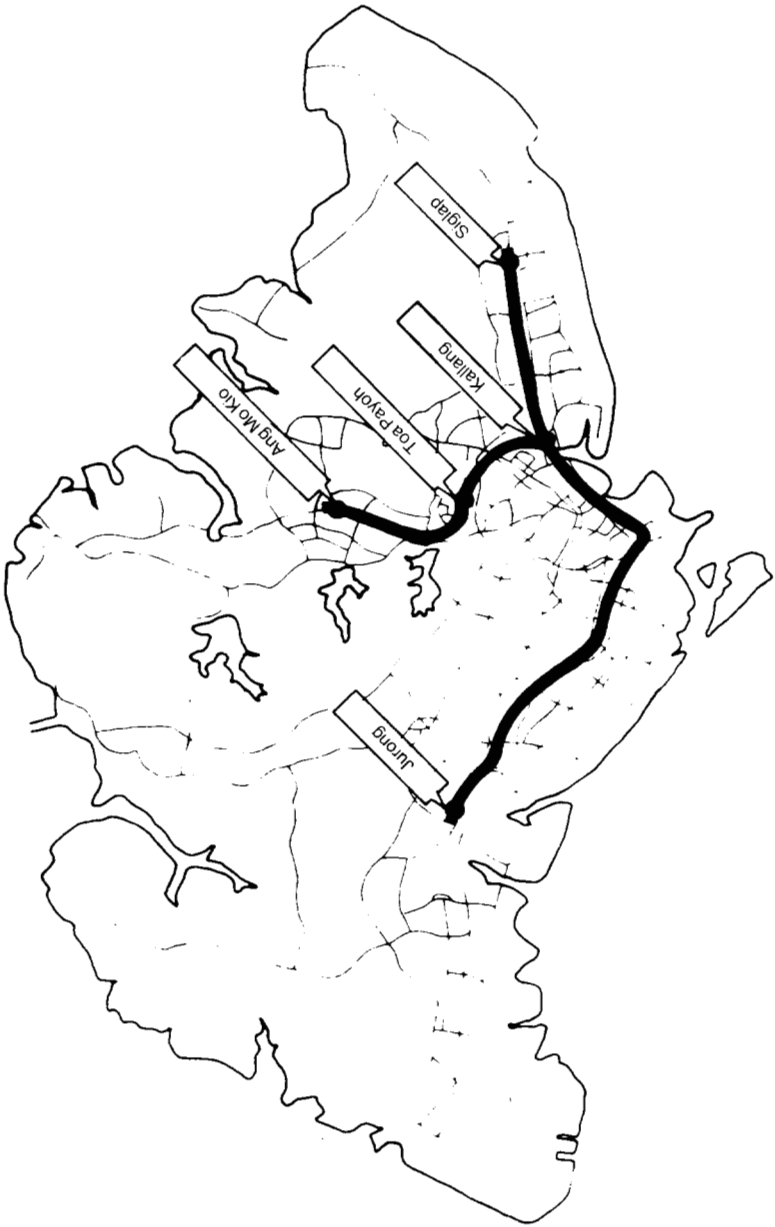


Fig. 2. MRT route proposed by state and city planning study (1971)

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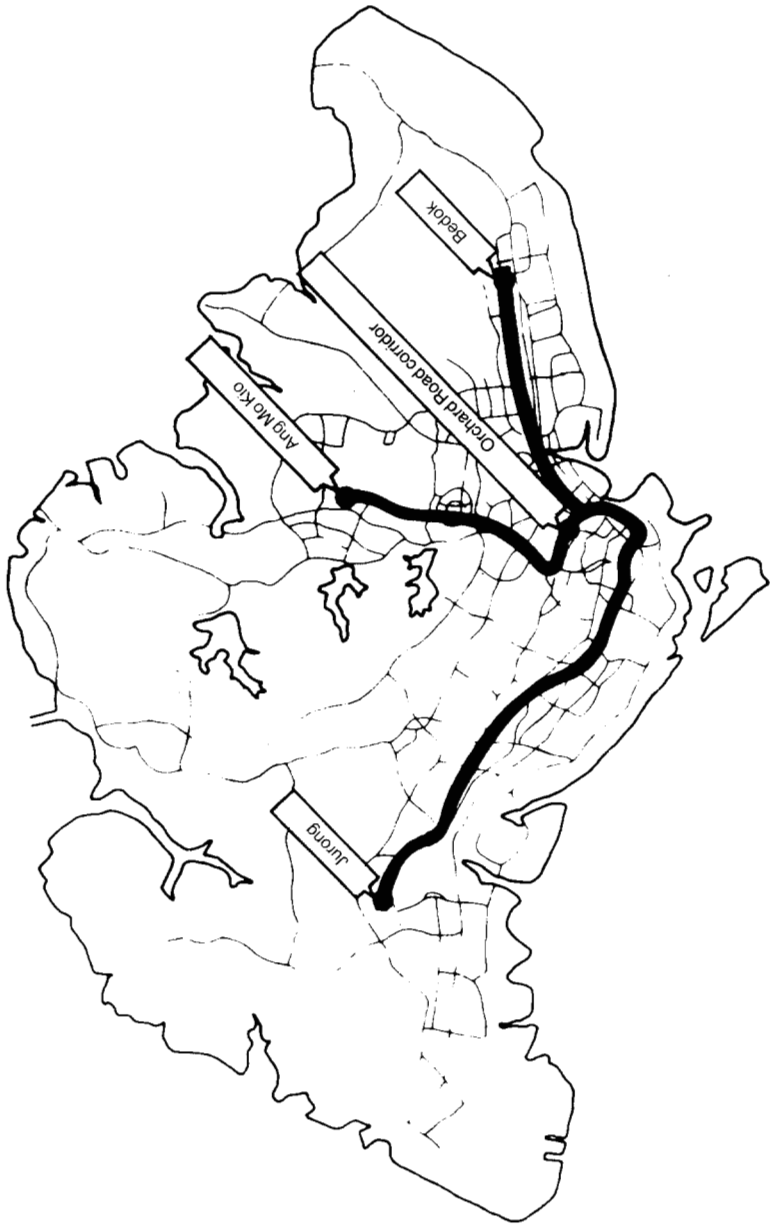


Fig. 3. MRT route proposed by mass transit study phase 1 (1974)

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Raffles Place). The preferred line was extended to the east, north and west to serve new public housing estates and industrial park areas. MTS phase III recommended lines very similar to those which have been or are being constructed (i.e. 42 stations over a 67 km route, one third of which is underground) (Fig. 4).

12. In 1980, increasing public awareness of the MRT proposal led to a public debate over the need for a mass rapid transit system for Singapore. The debate and the questions raised therein prompted the Ministry of Finance to commission a review team from Harvard University to take a fresh look at the topic of public transport in Singapore. The review team reported that an all-bus system could serve Singapore's future transport needs as well as an MRT railway and at a much lower cost. The all-bus scheme was essentially a 'hub and spoke' system, whereby feeder and express bus services would be linked and co-ordinated to enable commuters from the many public housing satellite towns to be transported to the central area.

13. In 1981, as a result of the Harvard report, the Government set up the Provisional Mass Rapid Transit Authority (PMRTA) which commissioned a comprehensive study to examine the feasibility of the all-bus proposal. This study, known as the comprehensive traffic study (CTS), was conducted by international transportation consultants (Table 1). The CTS concluded that, while the all-bus system could be made to work, the resulting traffic management measures and environmental impact would be so severe as to be unacceptable. A rail system was again seen as the better option.

14. In May 1982, soon after the conclusion of the CTS, the Government announced its decision to implement the MRT rail system. Almost immediately after the green light was given, the PMRTA initiated the system design and project staging (SDPS) study. Several of the consultants who had been involved in the earlier MTS and CTS studies were engaged to update the phase III design and to finalize alignments so that specifications for the initial civil design contracts could be prepared and tenders invited as early as possible.

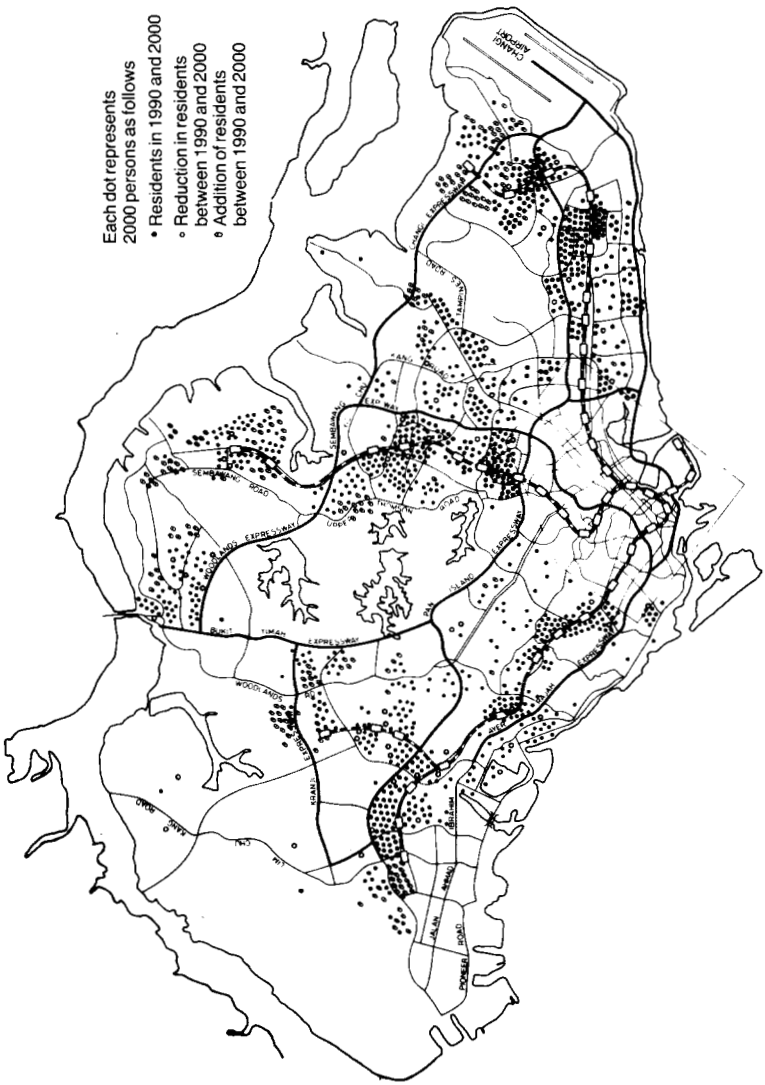
15. The PMRTA, in an effort to expedite implementation, decided to proceed with the north–south line, which was mostly underground, as the first phase of construction. This sequence was contrary to the recommendations in MTS phase III, and necessitated relocation of the main depot from the east–west line to the north–south line, since this had to be available at the start of revenue service. The rationale behind this decision was that the north–south line was predominantly underground and, on a linear cost basis, the more expensive line to construct. By implementing this line first, the Government would maximize benefits from the competitive prices which were anticipated from a depressed and eager construction market. A second factor was that the existing north–south corridor was poorly served by the road network and rush-hour congestion was acute. Early operation of the north–south MRT line would alleviate this problem and public utility would be best served.

16. In the SDPS study, the locations and alignments of a number of stations and depots, for which alternative locations, configurations and alignments had been developed during MTS phase III, were reviewed and finalized. The optimum location for depots and stabling sidings is just outside the central urban area so that trains can be introduced and withdrawn rapidly from the most intensively served sections of the lines. The sites chosen for Bishan main depot, Ulu Pandan sidings and Changi depot are all so located, and are therefore in areas where land values are decreasing. The additional costs incurred by commercial and residential

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Fig. 4 (above and below). MRT route proposed by mass transit study phase III: (a) generally adopted by the system design and project staging study (1980); (b) super-imposed on population patterns for 1990 and 2000

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(b)

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developments which have to be supported over, and isolated from, the railway beneath made such developments unattractive compared with adjoining unrestricted locations. The railway is not allowed to benefit from property developments, these being viewed as the prerogative of other designated government bodies—the Urban Redevelopment Authority in the central area, the Housing and Development Board and the Jurong Town Council (commercial developments) elsewhere.

Political and financial considerations

17. The decision to proceed with the MRT followed after a decade of studies and debate. For such a period to elapse between initial studies and implementation was not considered unreasonable when compared with the experience on similar projects elsewhere. Furthermore, given the scale of resources required for the MRT project and the circumstances of Singapore at the time, it was not surprising that the Government regarded the development of an MRT system as less urgent than the provision of health care, education, public housing and other infrastructure required to spur economic growth.

18. For much of the last two decades, Singapore has experienced high economic growth rates. This made it possible for the Government to accumulate sufficient financial reserves for the MRT project to be voted as part of the Government's infrastructure development budget. The various calls on this budget meant that a maximum of 5% could be allocated to the MRT project in any one year. The construction programme was therefore drawn up to meet this rate of expenditure. This direct funding out of government revenue allowed Singapore to select the optimum timing for the project and greatly simplified its implementation. It also avoided any need to accept financial offers from tenderers unless these were clearly more rewarding than the returns obtained by the Singapore Government in its own investments. No other funding arrangements were required except for the forward purchase, by the Monetary Authority of Singapore, of any foreign currency obligations entered into by the Corporation as part of a financial offer by a successful tenderer, in order to minimize currency fluctuations.

19. Since the early 1960s, Singapore has been carrying out land reclamation wherever feasible along the island's foreshore as a deliberate policy to increase land resources. As a result, sizeable plots of reclaimed land are now available. An important tract of this new reclamation is Marina South, located directly south of the present central business district. The possibility that the MRT system could serve Marina South formed an important element in the decision to implement the project. It was argued that with the accessibility provided by the proposed rail system, the commercial value of the reclaimed land would be increased significantly. It was estimated that, while the Government would finance the initial construction of the MRT, the enhanced value of development land in Marina South, when eventually realized, would be more than sufficient to offset the capital cost of the MRT.

20. The timing of the decision to build the MRT was, to a significant extent, influenced by the economic downturn in Singapore and elsewhere in the early 1980s. As a consequence, construction prices weakened and became very competitive. The recession provided an important reason for the Government to proceed with the MRT system, not only as a measure to boost a declining economy but also to take advantage of a very competitive construction market. The wisdom of

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this decision was borne out by the tender prices of the initial civil engineering contracts.

21. Subsidized public transport is a common and accepted policy in many developed countries. In Singapore, the Government, known for its policies of encouraging self-sufficiency, made clear that, although it would finance the initial capital cost of the MRT system, the railway operation was to be completely self-supporting in that revenue should be sufficient to cover all operating, maintenance and replacement costs. Therefore, the system was to be designed with operating efficiency as a paramount objective without causing undue detriment to the environment. In 1987, in accordance with government policy, a new public company, the Singapore Mass Rapid Transit Ltd, was formed, which has effectively leased the railway premises and is responsible for system operation and maintenance and replacement of the assets. It is anticipated that this company, currently wholly Government owned, will be offered for sale to the public once favourable financial performance has been achieved.

22. While phases I and II of the MTS were concerned essentially with establishing the concepts and criteria of the transport system, the order of cost estimates for the proposed Singapore system were established from unit costs derived from systems in other countries. Such a method of cost forecasting, although acceptable in the initial stages, lacked finesse and local colour and was likely to be inaccurate. One of the principal objectives of phase III of the MTS was to establish a reliable cost estimate for the design, construction, equipping, testing and commissioning of the proposed system.

23. The phase III estimates were prepared on the assumption that construction contracts would be awarded on the basis of the traditional system of detailed engineer's designs with bills of quantities and with the tenderers selected from among contractors with experience of similar works. It was further assumed that station building and finishings works would be carried out by local contractors as sub-contractors to the main contractors; the intention was to bring benefit to local industry from the technology transfer of having experienced international contractors co-ordinate the contract activities.

24. A key element in the phase III estimates was the assessment of realistic tunnelling rates of advance. Table 2 indicates the forecast and actual rates.

25. To ensure accuracy the estimates were built up empirically from labour, plant and material, with the inclusion of lump sum prices kept to a minimum. Labour rates were based on information provided by local contractors. Material costs were derived from current quotations, and plant rates included both the capital and running costs. Allowance for contingencies, always a matter of subjection, was based on the various work categories (Table 3). An allowance of $10\frac{1}{4}\%$ was included to cover the cost of design and supervision. Land costs were established, but only those relating to the acquisition of private land were included in the cost estimate. Use of state land, where required, was valued but not charged against the mass transit railway estimate.

26. The final cost estimate of S$3950 million (£823 million) at December 1979 prices was related to the system shown in Fig. 5. The estimate was sectionalized to facilitate appraisal by the Government.

27. When the final decision to proceed with the MRT rail system was made in 1982, the approved alignment differed from phase III in that the extension to Changi Airport had been omitted and a number of minor revisions had been made. Additional costs for the strengthening of nine underground stations to civil

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Table 2. Forecast of tunnelling rates and actual rates

Section	Forecast: m/week	Actual: m/week	Comments
Tiong Bahru to Tanjong Pagar	30	20·6	Ground fracturing found to be more extensive than anticipated and face support requirement increased.
Tanjong Pagar to Raffles Place	10·8	16·8	Estimate allowed for use of compressed air only. In practice, the whole of the marine clay length (500 m approximately) was treated by jet grouting.
Singapore River to City Hall	15	19·3	
City Hall to Dhoby Ghaut	Assumed cut and cover construction	15·3	
Dhoby Ghaut to Somerset	23	26·3	
Somerset to Orchard	30	31·2	Higher tunnelling speeds resulted from use of one fully mechanized shield for all tunnels. Overall construction time approximately equivalent to two Greathead type shield drives assumed in estimate.
Orchard to Newton	20	50·7	
Newton to Novena	15	13·6	
Novena to Toa Payoh Southward from Shan Road	29·4	10·2	Overall progress of NATM drives reduced by a 120 m section in decomposed granite excavated under compressed air by electric backhoe.
Novena to Toa Payoh Northward from Shan Road	15	8·7	Unforeseen 90 m length of full face of rock slowed shield. Inadequate bulkhead and air locks had to be modified during the drive to permit use of compressed air pressures in excess of one bar.

defence standards had also to be included, as had provision for the Transit Authority's office, Operations Training School, and so on.

28. After allowance for the system modification and after updating the December 1979 estimate by using construction indices for the period 1979/81 for civil and E&M works, a target figure of S$5000 million was established as the project budget. The estimated out-turn cost at the beginning of 1989 is S$5050 million

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Table 3. Contingency allowances

Work categories	%
Elevated sections between stations	5
Elevated stations	5
At grade construction	5
Transitions	10
Underground cut and cover between stations	10
Underground stations	10
Driven tunnel	15
Trackwork	20

(£1357 million) which, expressed in terms of 1982 dollars for comparison, is equal to a range of S$4500 million to S$4700 million depending on the assessment of inflation. (The exchange rate between the S$ and the £ varied from a low of S$1 = £0·20 in 1980 to a high of S$1 = £0·39 in 1984.)

29. Typical costs to date are as follows.

(*a*) *Above ground.* The twin track viaduct cost per metre has varied between S$5970 to S$7530 with an average of S$6610. The basic structure costs for the twin track single island platform elevated stations averaged S$4·3 million, while those for the three track two island platform stations averaged S$6·7 million. The finishes costs were approximately equal to the basic structure costs.

(*b*) *Below ground.* These costs fluctuated greatly, reflecting the rapidly varying ground conditions in Singapore. The bored tunnelling rates varied between S$5710 and S$19 880 per metre. The costs of cut and cover twin box varied between S$29 500 and S$38 140 per metre, but for special structures such as the quadruple cell crossing of the Singapore River, the cost rose to approximately S$200 000 per metre. The underground station costs varied between S$18 million and S$24·5 million for a simple two level twin track basic structure, depending on location. Civil defence requirements, the provision of commercial levels, etc., caused costs to fluctuate greatly in other stations. The average twin track underground station basic structure cost was S$33 million, with finishes averaging S$12·5 million per station.

Planning controls and route protection

30. The relative ease, in terms of integration with existing development, with which the MRT has been and is being implemented was due largely to the early steps taken to safeguard the MRT routes. The adoption by Government agencies of the state and city planning (SCP) development concept land-use plan as a basis for the control of physical development in Singapore was fundamental and essential. Safeguarding the various MRT routes was carried out very simply by checking for encroachment into the defined zone of protection. This was done with the assistance of two agencies, both under the control of the Ministry of National Development and both empowered to impose enforceable restrictions on development proposals: the Planning Department was responsible for public sector development control; the Development and Building Control Department was

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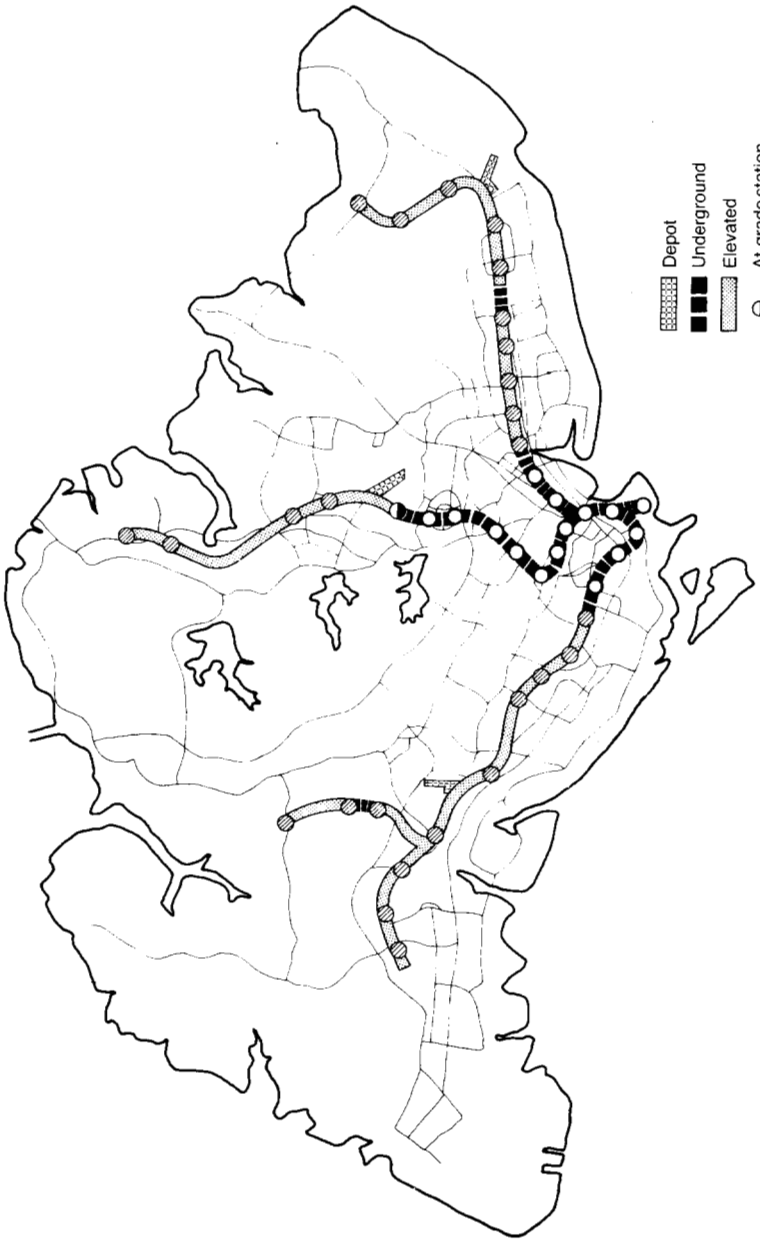


Fig. 5. System under construction showing at-grade, underground and elevated sections

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responsible for controls in the private sector. An agreed procedure was established by which development proposals were referred to the Mass Transit Study Unit if a proposed development fell within the zone of protection (which was defined as being within 100 m of the proposed MRT). In some cases, development was allowed to proceed subject to the imposition of conditions requiring developers to make provision for the integration of their building and the MRT.

31. In the terms of reference for MTS phase III, a specific objective was included which required the consultants to develop and refine the guidelines and criteria being used to safeguard the MRT route. Due account was to be taken of the technical and environmental aspects related to the type of proposed development. Two developments, the Shell Tower and Chartered Bank, were referred to the Mass Transit Study Unit and, as a result, the foundations of both buildings were modified to facilitate later MRT construction.

The route

The route in general

32. By the time the decision to construct the MRT system was taken, the land along the route of the system was relatively well developed and that in the city area very heavily so. The dominating visual effect of an all-viaduct system through the business, commercial and governmental centres of Singapore was not acceptable in the environmentally conscious Republic. Allied to this was the economic sterilization of valuable land by stations and viaduct supports.

33. In the case of underground railways, only the entrances to stations occupy prime sites and these can often be incorporated into the existing infrastructure or into future commercial developments. Construction directly over the tunnels usually involves increased expenditure on foundations, but in areas of high-cost land, such as the Singapore central business district, the increase forms a relatively small proportion of the building cost and does not inhibit development. With these factors in mind, planning for the Singapore MRT system placed the railway underground throughout the central area, emerging above ground at the earliest opportunity, consistent with the density and type of development.

34. Outside the central area the line was based on the 'string of pearls' philosophy, linking the various new towns and industrial park developments. To avoid major disruptions to the road systems the railway had therefore to be predominantly elevated. It was only in a few intermediate sections that an at-grade alignment could be considered.

35. To maximize the benefit to be obtained from the MRT, provision for integrated commercial development has been made at many of the underground stations. Convenient connections are provided to adjoining buildings and facilities (Fig. 6); elsewhere, the present entrances are temporary and will be removed or adapted to allow integration into future developments, e.g. Newton, Somerset, Bugis and Lavender.

36. Where stations underlie the sites for future development, the structural box has been designed to support a podium type structure (4–6 levels with commercial floor loadings of 2·5 kn/m^2 to 5 kn/m^2) with the adjoining areas of the development site accommodating high rise structures (Novena, Orchard, Dhoby Ghaut, Tanjong Pagar and Tiong Bahru). These technical requirements will be incorporated in the development criteria as and when the Urban Redevelopment Authority, the statutory body which regulates the sales and use of government land in the central area, invites tenders for these sites from private developers.

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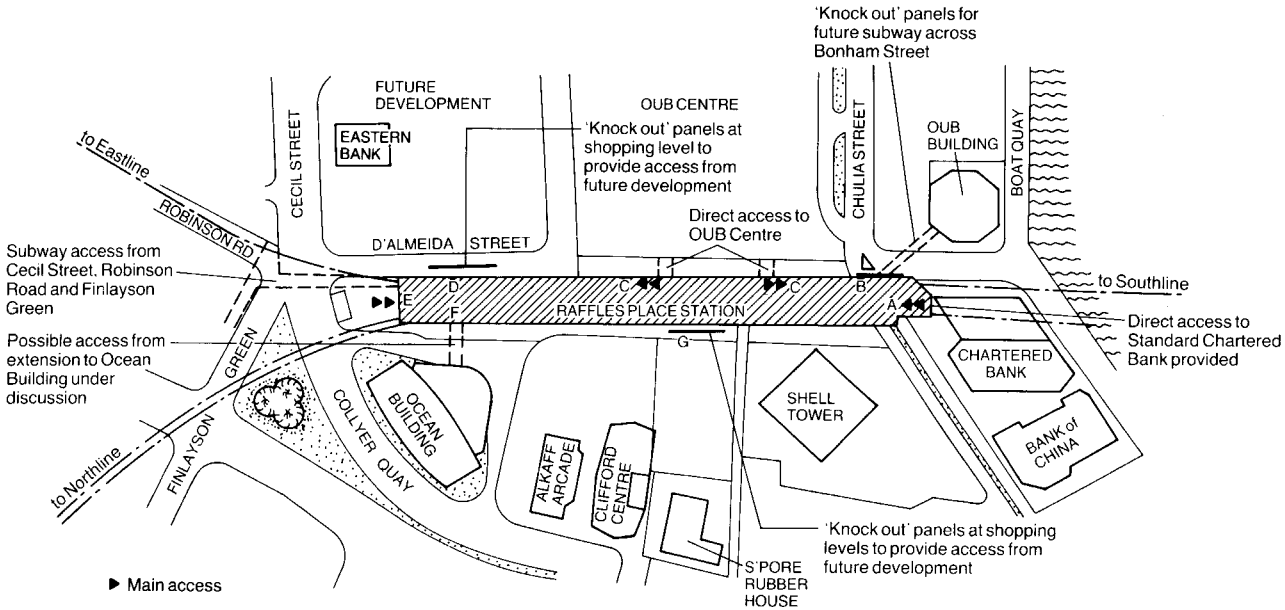


Fig. 6. Relationship of Raffles Place Station with adjacent existing and proposed developments

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37. In the outer areas many of the stations are located in the new town centres with convenient bus interchange facilities to widen the catchment area.

The route in particular

38. The western terminal station at Boon Lay is located alongside the proposed Jurong West bus station. This will ensure convenient interchange with feeder buses to the Jurong industrial estate. From here the line passes through Lakeside and Chinese Garden serving residential and recreational areas.

39. Jurong East station is set in the middle of the new town, the design being integrated with a bus station. The branch line serving the new towns of Bukit Batok, Bukit Gombak and Choa Chu Kang leaves by a grade separated junction west of the station. The station itself has two island platforms serving three tracks. The centre track facilitates train reversing as well as movements to and from the depot by a grade-separated junction. A further depot branch accesses the main line towards the east.

40. Clementi station is in the new town centre with easy access to the town bus station. Between Clementi and Buona Vista there is little traffic potential and this is the longest (2·7 km) interstation section on the east–west line. Buona Vista station is located adjacent to a proposed bus station to serve many nearby residential and educational establishments.

41. The line runs through Commonwealth, Queenstown and Redhill stations serving mixed industrial/residential areas. East of Redhill, on the fringe of the central business district, the line goes underground to Tiong Bahru and Outram Park stations. The next station, Tanjong Pagar, is at the south end of the central business district. Raffles Place lies in the heart of the Singapore business and financial centres. From here, the line crosses the Singapore River to City Hall station serving governmental and commercial areas.

42. Leaving City Hall, the line follows one of Singapore's main traffic arteries from the east. Bugis and Lavender stations serve the surrounding mixed residential and commercial areas. The line then makes a transition to viaduct where it crosses the Kallang River and runs along a cleared corridor through residential, commercial and light industrial developments. Kallang, Aljunied, Paya Lebar, Eunos and Kembagan stations serve this area. A short length of tunnel brings the line into the central median of the road with Bedock and Tanah Merah stations serving Bedok New Town. Tanah Merah station has three tracks to give grade-separated access to Changi Depot. The main line continues (incorporating provision for a future grade-separated branch to Changi Airport) to Simei, Tampines (with an adjacent bus station) and on to the eastern terminal station at Pasir Ris; each of these stations serves a new town of the same name.

43. From the city area the line north passes through Raffles Place and City Hall stations, running parallel with the east–west line. It then turns to follow the Orchard Road tourist area and serves the associated entertainment centres, hotels and shops. Dhoby Ghaut, Somerset, Orchard and Newton stations provide convenient access.

44. Toa Payoh station is located beneath the major bus station in the centre of the original new town of Toa Payoh. Covered interchange facilities are provided. Braddell station serves the new town's northern extension. The tracks emerge from tunnel into cutting, the semi-depressed Bishan station serving the new town of the same name. Ang Mo Kio station, with a subway to the bus station, serves this very large new town. The line then runs through Yio Chu Kang and Khatib to its

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present termination at Yishun, each station serving large residential areas. The overrun tracks will permit further extension north if required.

45. From the city southwards, Marina Bay station lies in a relatively open area scheduled for major commercial development. The overrun tunnels south of the terminal station are to facilitate further extensions into Marina South and possibly across the mouth of the Singapore River to the reclaimed land on the east side.

Implementation

Staged construction and opening

46. As the MRT system has a route length of 67 km and 42 stations, it was essential to schedule construction in a number of sections. The basic criterion was that, as far as reasonably possible, each section would be economically viable from an operational point of view. (Revenue should exceed operating costs plus amortized replacement costs.) Nevertheless it was necessary to recognize particular traffic conditions and construction and operating restraints.

47. An essential facility for an operating railway is a depot for storage and maintenance. In a small island, land use is of relatively greater importance than in many states and, therefore, the 30 hectare land area required for the main depot represented a significant commitment for Singapore—particularly when efficient railway operation requires the depot to be on the fringe of the inner urban areas. Before the decision was made to construct the north–south line first, careful review had identified an adequate area of land adjacent to the proposed Bishan New Town and alongside the Kallang River. The land was low lying, marshy and formed part of the flood plain of the river. Although the land is strategically placed, these factors had prevented the site from being earmarked for development. The expense of using this land as a railway depot was considerably higher than that of using other locations which were examined. Nevertheless, the overall advantages to the state were considered to outweigh the cost disadvantages and the decision was made to adopt Bishan as the site for the main depot. This effectively determined the construction phases of the railway. (Fig. 7).

48. The first phase encompassed the depot and joined the major new town of Ang Mo Kio to the central business district. Outram Park station, where permanent reversing facilities were required, provided a convenient temporary termination point. The north–south and east–west lines were both constructed between City Hall and Raffles Place. This represented the majority of the north–south flow. However, this initial section serves traffic flows only to and from the central business district, a further commuter flow being from the north towards the major industrial complex in the west at Jurong. To accommodate this, an extension westwards, namely phase 1A, was required to follow as rapidly as possible. This extension comprises the east–west line to Clementi, including the Ulu Pandan stabling sidings.

49. The very competitive nature of the bidding for civil construction work world-wide in 1983–84 was clearly reflected in the successful bids for phases 1 and 1A. Aware that phase 2 had to go ahead if the MRT project were to fulfil the country's expectations, the Government readily accepted the economic argument for advancing construction of phase 2 to benefit from the possibly limited period of recession. The MRT Corporation, which in 1983 succeeded the Provisional Authority, reviewed its workload and was able to advise the Government that bids for the construction of phase 2 could be invited eight months after the decision to

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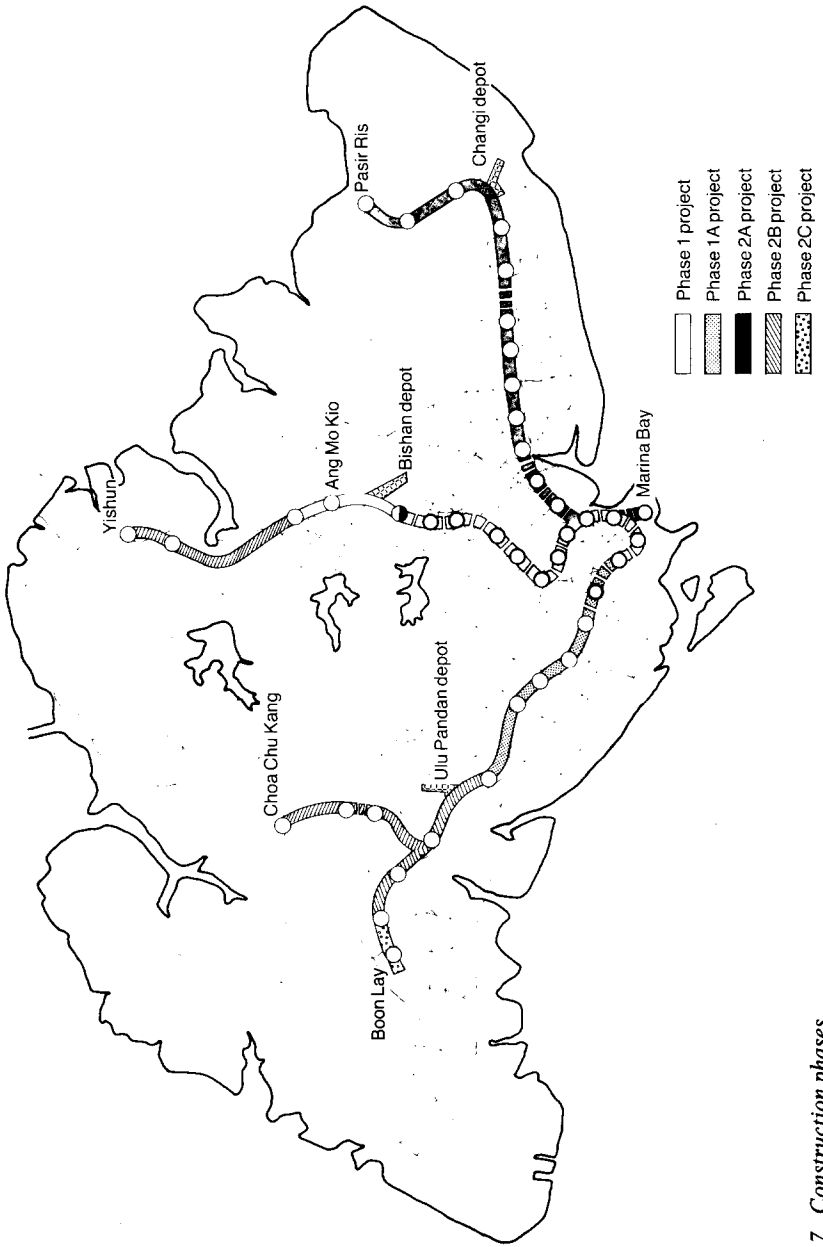


Fig. 7. Construction phases

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proceed was given. Inherent in this decision was the awareness that contract drawings and documents, while adequate to invite competitive bids, would require considerable post-award refinement.

50. Furthermore, instead of a gradual transfer of its own supervision and management staff from phase 1 to phase 2, the Corporation would be required to increase an already massive recruitment programme, particularly for site staff, so that implementation of all phases could be supervised simultaneously. Despite these drawbacks, the advantages stemming from the eagerness of contractors to bid competitively and from the quicker utilization of the complete MRT system were recognized and the Corporation received approval in 1985 to proceed with phase 2. The objective was to have the whole system operational by 1990, seven years after the start of construction and two years earlier than the originally programmed completion date of 1992.

51. Phase 2A comprised the completion of the southern extension of the north–south line to Marina Bay and the completion of the eastern arm of the east–west line to Pasir Ris, with provision for extension to Changi Airport if required. Changi sub-depot was included as part of this line (Fig. 7). Phase 2B comprised the extension of the east–west line west to Lakeside, the branch to Chao Chu Kang and the northern termination of the north–south line at Yishun (Fig. 7). Phase 2C is the final westward extension to Boon Lay.

52. Within these constructional phases are several sub-sections which have opened or will open to revenue service in the stages shown in Fig. 8.

Future plans

53. When the first two lines were under construction, the Corporation undertook the review of a possible extension of the railway system to the north-east sector of the island, which already contains industrial and commercial estates and the major new town of Hougang. Other projected new towns in the north of the island at Punggol and Jalan Kayu would fall within the catchment of this line. The core of a new line, however, was to be the Serangoon corridor which, since 1968, had been recognized as a vital traffic objective. The North-East Sector Study reported in late 1985. It recommended an alignment starting from Outram Park, with an interchange with the east–west line, through Peoples Park to an interchange with the north–south line at Dhoby Ghaut, and then following Serangoon Road to Hougang with branches to Punggol and Jalan Kayu (Fig. 9). The study presented an economic case which indicated that the line would not be required until very late in the century and that to proceed immediately with the construction of the new line, so as to permit continuity of construction, would lead to an unnecessarily early commitment of resources against a number of years of low revenue return.

54. As Singapore continues to develop, the Corporation is constantly reviewing the future configuration of the railway system to ensure optimum services. Recent changes in the demographic characteristics of the country have prompted a radical review of housing/new town philosophies, which has resulted in a shift towards more intensive development of existing infrastructure rather than the creation of new towns. In line with this view, the Corporation is currently examining an extension to the expanding town of Woodlands (Fig. 10).

55. Phase 1, sections 1 and 2, and phase 1A were opened 7 November 1987, 12 December 1987 and 12 March 1988 respectively. Phase 2, sections 4 and 5, was

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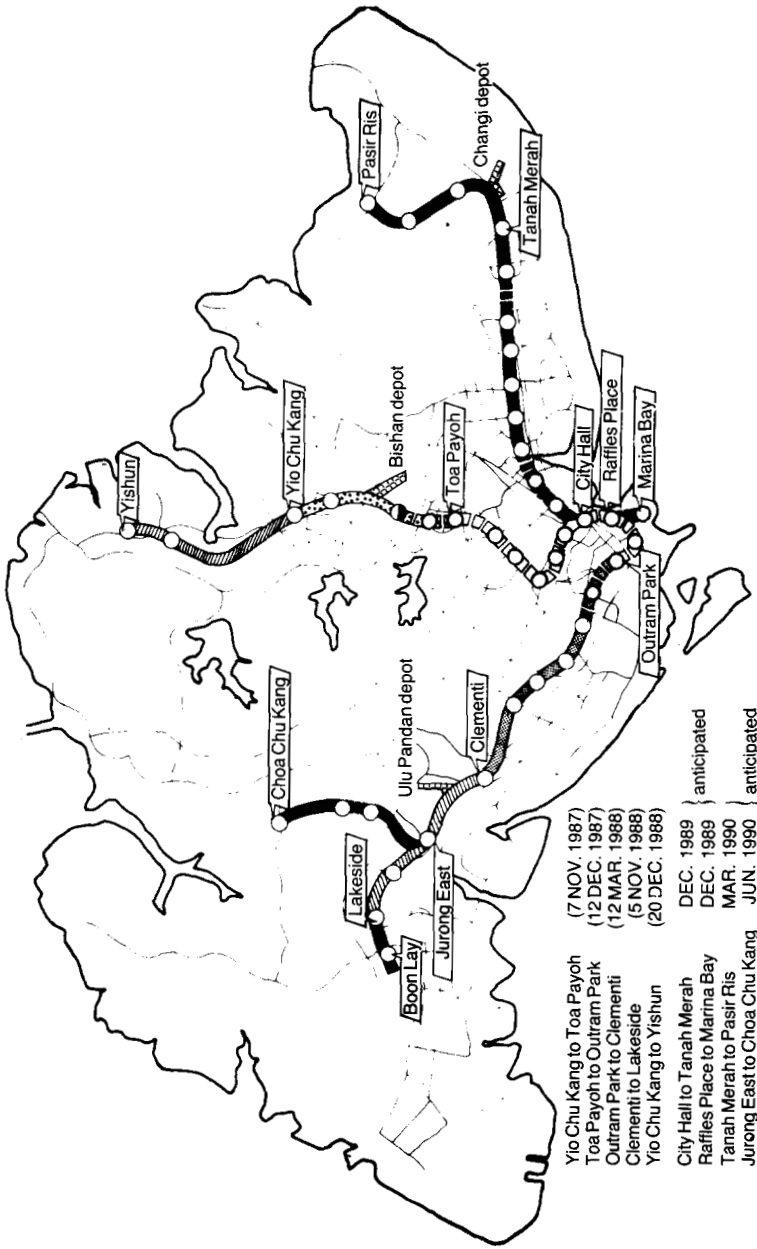


Fig. 8. Stages for start of public opening

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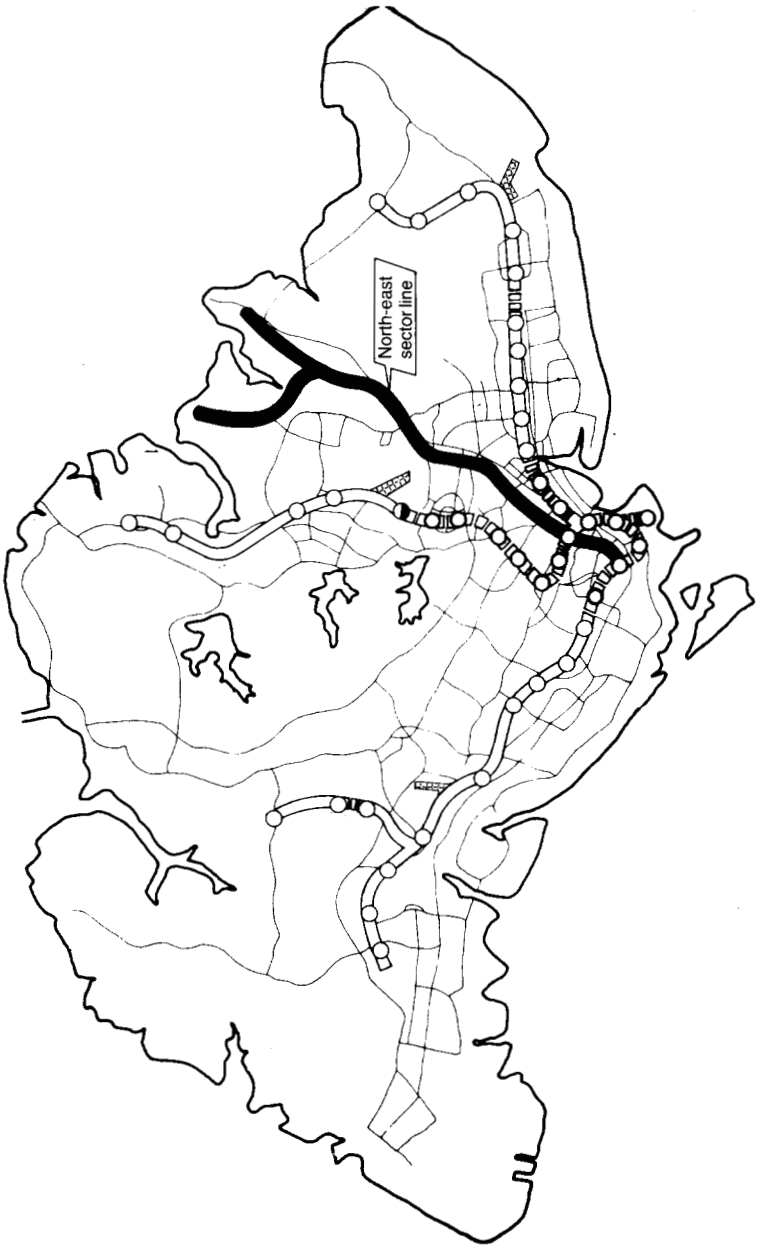


Fig. 9. Proposed alignment for north-east sector line

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Fig. 10. Proposed alignment for Woodlands extension

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opened on 5 November 1988 and 20 December 1988 respectively; all openings have been marginally ahead of programme. All other sections remain on schedule.

Contractual aspects

Contract arrangements

56. When the Singapore Government gave approval for the project to proceed, the client (initially, the Provisional Mass Rapid Transit Authority, and subsequently, the Mass Rapid Transit Corporation) wanted the earliest possible start of revenue service compatible with a technologically advanced and aesthetically pleasing railway system.

57. The civil engineering construction industry in the South-East Asia region was in recession in 1983, and an early award of construction contracts offered the advantage of strong competition and lower prices. Several international tunnelling contractors were completing work on the Hong Kong Mass Transit Railway and were well placed to move into Singapore.

58. The prospect of a turnkey contract for the whole of the project was considered briefly and rejected. Such an arrangement would not necessarily have resulted in lower prices or an accelerated programme. Furthermore, the risk involved in putting total reliance on a single contractor (albeit possibly comprising several joint venture partners) was not considered to be in the best interests of this nationally important project.

59. A multi-contract arrangement was chosen, whereby civil works on phase 1 were split initially into eight civil engineering contracts. Separate contracts were to be awarded for the MRT main depot at Bishan and for the tracklaying works.

60. Of the 14 stations proposed for the 20 km phase 1, two were elevated, one in a cutting and the remaining 11 were underground.

61. In 1983, the Government directed that five of the phase I underground stations were to incorporate civil defence facilities. Conceptual designs needed to be amended to incorporate gas-tight and blast-proof installations. To avoid delay to the start of the works, the proposed eight civil engineering contracts were allowed to proceed after the civil defence stations had been omitted. Five separate civil defence station contracts were subsequently awarded. In all, 15 civil engineering contracts were awarded for phase 1 (see Table 4).

62. Phase 1A comprises five overhead and one underground (civil defence) stations. The track is carried on a viaduct for almost the whole length of 9 km. Five civil engineering contracts were awarded for this phase, one of which was an advance earthworks contract at the Ulu Pandan sub-depot site.

63. Phases 2A (19 km) and 2B (18 km) comprise 19 overhead and three underground (civil defence) stations. The track is carried mostly on a viaduct with tunnels adjacent to and between the underground stations. Fourteen phase 2 civil engineering contracts were awarded, one of which was an advanced earthworks contract at Changi sub-depot.

64. The phase 1 trackwork contract was extended to cover trackwork requirements on phases 1A, 2A and 2B.

65. Singaporean civil engineering contractors possessed neither the specialist expertise nor adequate resources to undertake the MRT works without overseas support. Few major tunnels had been driven in the Republic and ground conditions along much of the underground route were known to be difficult. Consequently, the need to attract overseas contractors who could offer mass transit and tunnelling experience was recognized as essential.

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Table 4. Particulars of main civil and structural contracts

Phase	Contract	Contractor	Description		(a) Date of award (b) Contract completion date	Awarded contract sum (including provisional sums)
1	101	Nishimatsu Construction Co/ Lum Chang Bldg Constr	O/L*	Construction of Bishan depot	(a) 10.09.84 (b) 25.05.87	S$164 412 000·00
1	102	Paul Y Construction Pte Ltd	O	Construction of Yio Chu Kang and Ang Mo Kio (elevated) Stations' viaduct sections (3.7 km)	(a) 08.10.84 (b) 23.04.87	S$82 655 654·43
1	103	Sinbelco Construction Co	O/L	Design and construction of Bishan (at ground) Station, tunnels between Bishan and Braddell (1.52 km)	(a) 14.12.83 (b) 02.01.87	S$32 882 507·00
1	103A	Sinbelco Construction Co	O/L	Design and construction of Braddell (underground) Station (civil defence), tunnels between Braddell and Toa Pao (0·89 km)	(a) 09.04.84 (b) 27.04.87	S$ 58 078 939·00
1	104	Tobishima/Takenaka JV	O	Design and construction of Toa Payoh and Novena (underground) Stations, tunnels between Toa Payoh and Newton (2·61 km)	(a) 14.09.83 (b) 08.01.87	S$96 802 000·00
1	105	Bocotra Construction Pte Ltd	O/L	Design and construction of Orchard (underground) Station, tunnels between Newton and Somerset (197 km)	(a) 03.10.83 (b) 21.03.87	S$78 800 000·00
1	105A	Dragages–Sembawang Construction Pte Ltd	O/L	Design and construction of Newton (underground) Station (civil defence) adjacent (0·27 km) cross-over section (0·27 km)	(a) 09.01.84 (b) 22.06.87	S$43 462 348·00

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Table 4—continued

Phase	Contract	Contractor	Description		(a) Date of award (b) Contract completion date	Awarded contract sum (including provisional sums)
1	106	Campenon Bernard/ Singapore Piling & Civil Engineering Co Ltd	O/L	Design and construction of Dhoby Ghaut Station (underground), tunnels between Somerset Station and City Hall (1·50 km)	(a) 10.10.83 (b) 23.06.87	S$51 300 000·00 and FFr31 150 000·00
1	106A	Bocotra Construction Pte Ltd	O/L	Design and construction of Somerset (underground) Station (civil defence)	(a) 26.01.84 (b) 22.06.87	S$39 257 210·00
1	107	Kajima/Keppel JV	O/L	Design and construction of tunnels between City Hall and Raffles Place (0·81 km) NB/SB (0·81 km) EB/WB	(a) 10.10.83 (b) 23.06.87	S$35 650 000·00
1	107A	Taisei–Shimizu–Marubeni JV	O	Design and construction of Raffles Place (underground) Station (civil defence); tunnels towards Marina Bay (0·43 km) NB/SB (0·29 km) EB/WB	(a) 07.05.84 (b) 22.06.87	S$70 723 000·00
1	107B	Nishimatsu Construction/ Lum Chang Bldg Construction	O/L	Design and construction of City Hall (underground) Station (civil defence), tunnels towards Bugis (0·28 km) NB/SB (0·48 km) EB/WB	(a) 07.05.84 (b) 22.06.87	S$77 655 600·00
1	108	Taisei–Shimizu– Marubeni JV	O	Design and construction of Tanjong Pagar (underground) Station tunnels between Raffles Place and Tanjong Pagar (1·12 km)	(a) 14.12.83 (b) 17.05.87	S$63 387 500·00

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1	109	Ohbayashi/Okumura JV	O	Design and construction of Outram Park (underground) Station tunnels between Tanjong Pagar and Tiong Bahru (2·32 km)	(a) 02.11.83 (b) 27.05.87	S$73 850 000·00
All	110	Henry Boot/Gammon/ Singa JV	O/L	Trackwork for phase I, phase IA and phase II (option)	(a) 07.01.85 (b) 26.12.88	£53 569 000·00 and US$59 148 000·00 and S$3 477 450·00
1A	201	Bocotra Construction Pte Ltd	O/L	Construction of Tiong Bahru (underground) Station (civil defence), cut and cover tunnels between Tiong Bahru and headwall (0·96 km)	(a) 08.04.85 (b) 30.11.87	US$13 724 411·00 and FFr25 028 333·00
1A	202	Sumitomo Construction Co Ltd	O	Construction of Commonwealth and Redhill (elevated) Stations, viaduct between headwall and Queenstown Circus (2·51 km)	(a) 11.02.85 (b) 30.11.87	S$50 767 979·46
1A	203	Aoki/Lim Kah Ngam JV	O/L	Construction of Queenstown and Buona Vista (elevated) Stations, viaduct between Queenstown Circus and Clementi Road (3·61 km)	(a) 07.01.85 (b) 30.11.87	US$32 508 375·00
1A	204	Aoki/Lim Kah Ngam JV	O/L	Construction of Clementi (elevated) Station, viaduct between Clementi Road and Ulu Pandan Depot (2·24 km)	(a) 22.04.85 (b) 30.11.87	US$21 900 000·00
1A	205	Hiap Shing Construction Pte Ltd	L	Ulu Pandan Depot (earthwork)	(a) 12.08.85 (b) 12.05.86	S$2 677 363·00

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Table 4—continued

Phase	Contract	Contractor		Description	(a) Date of award (b) Contract completion date	Awarded contract sum (including provisional sums)
2A	301	Nishimatsu Construction/Lum Chang Bldg Construction	O/L	Design and construction of Bugis and Lavender (underground) Stations (civil defence), tunnels between City Hall and Lavender (2·39 km)	(a) 14.10.85 (b) 25.09.89	S$227 112 437·50
2A	302	Lee Kim Tah Pte Ltd/ Societe Generale D'Enterprise Sainrapt Et Brice JV	O/L	Construction of Kallang and Aljunied (elevated) Stations, viaducts between Lavender and Paya Lebar (2·75 km)	(a) 12.11.85 (b) 25.09.89	S$59 520 783·39
2A	303	Okumura/Oh Teck Thye JV	O/L	Construction of Paya Lebar and Eunos (elevated) Stations, viaduct between Paya Lebar and Kembangan (2·26 km)	(a) 16.12.85 (b) 25.09.89	S$60 005 677·48
2A	304	RDC Pte Ltd/Ohbayashi Corporation	O/L	Construction of Kembangan and Bedok (elevated) Stations, viaducts between Kembangan and Tanah Merah (3·27 km)	(a) 13.01.86 (b) 25.09.89	S$35 048 866·00 and US$16 493 584·00
2A	305	Wah Chang Int'l Corporation (S) Pte Ltd/Hyundai Engineering & Construction Co Ltd	O/L	Construction of Tanah Merah (elevated) Station, viaduct between Tanah Merah and Changi Depot (1·95 km)	(a) 13.01.86 (b) 25.09.89	US$31 952 260·52
2A	306	Resources Development Corporation Pte Ltd/ Sato Kogyo Co Ltd	O/L	Construction of Simei, Tampines and Pasir Ris (elevated) Stations, viaduct between Tanah Merah and Pasir Ris (5·45 km)	(a) 10.03.86 (b) 30.10.89	S$91 891 290·00
2A	307	Gammon Ltd	O	Changi Depot Earthworks	(a) 07.10.85 (b) 07.12.87	S$9 969 496·50
2A	308	Jurong Engineering Ltd	L	Construction of Changi depot	(a) 08.12.86 (b) 30.10.89	S$27 472 514·75

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2A	310	Gammon (HK) Ltd/ Antara Koh Pte Ltd	O/L	Construction of Marina Bay (underground) Station (civil defence), tunnels between Raffles Place and Marina Bay (1·10 km)	(a) 07.04.86 (b) 25.09.89	S$96 247 606·00
2B	402	GTM Enterprise/GTM Int'l Societe Nouvelle Coignet Enterprise/Coignet SAJV	O	Construction of Yishun and Khatib (elevated) Stations, viaduct between Yio Chu Kang and Yishun (6·25 km)	(a) 12.11.85 (b) 09.01.89	S$92 555 379·16
2B	403	JDC Corporation/Jurong Engineering Ltd	O/L	Construction of Jurong East (elevated) Station, depot sidings and viaduct between Ulu Pandan and Jurong East (2·78 km)	(a) 07.10.85 (b) 31.10.88	S$79 580 169·33
2B	404	Aoki/LKN Corporation Pte Ltd	O/L	Construction of Chinese Garden and Lakeside (elevated) Stations, viaduct between Jurong East and Lakeside (3·03 km)	(a) 12.11.85 (b) 31.10.88	S$64 750 313·00
2B	405	RSEA Int'l Pte Ltd/Hock Lian Seng Engng Pte Ltd	O/L	Construction of Bukit Batok, Bukit Gombak and Chua Chu Kang (elevated) Stations, viaducts and tunnels between Jurong East and Chua Chu Kang (6·15 km)	(a) 06.01.86 (b) 05.02.90	S$50 688 995·20 and US$23 424 600·00
2C	501	RSEA Int'l Pte Ltd	O/L	Construction of Boon Lay (elevated) Station, viaduct between Lakeside & Boon Lay (1·48 km)	(a) 09.05.88 (b) 14.05.90	S$34 190 000·00
Misc	130	Wah Chang Int'l Corp Pte Ltd	L	Construction, completion and maintenance of MRTC headquarters building basement	(a) 11.04.88 (b) 08.01.89	S$6 789 000·00

* O is Overseas Contractor; L is Local Contractor

66. As a means of ensuring that local contractors were given an opportunity to compete and participate (thereby acquiring specialist know-how), financial incentives in the form of preferential margins applied to tender sums during tender evaluation were offered to local contractors tendering either solely or in joint venture. A preferential margin of up to 5% was offered to tenderers who could show a 50% local equity participation and maintain this throughout the contract period. The relief tapered with reductions in the amount of local equity. The preferential margin scheme went some way to offset the effect of export credit benefits which many overseas contractors were able to obtain from their national governments.

67. The majority of civil engineering contracts were awarded to joint ventures consisting of local and overseas contractors, performance guarantees being required from each joint venture partner. In many cases there was no necessity to apply the preferential margin as the successful tenderer submitted the lowest bid.

Contract divisions

68. Contract divisions were made with several criteria in mind. Contract size needed to be sufficiently large to attract international interest and to ensure economic use of costly or specialist plant and installations (tunnelling shields, pre-cast concrete viaduct beam casting yards, viaduct beam launching mechanisms).

69. Contract size was restricted to reduce the risk of any single contractor overstretching and thereby jeopardizing the project. Additionally, a large number of contracts was likely to result in greater participation from local contractors, a wider transfer of technology and more competitive prices.

70. Other factors influencing the contract split included geophysical boundaries (areas of hard and soft ground requiring different tunnelling methods or excavation techniques), changes in type of construction (underground/at-grade/overhead) and the availability of adequate worksites.

Conditions of contract

Historically, public works in Singapore have been contracted using forms of contract devised by government departments. Although these forms are well known to local contractors, they differ in their allocation of risks from the international forms and thus might have discouraged international tendering. Accordingly, the Corporation chose to seek tenders based on the International Conditions for Civil Engineering, the FIDIC Conditions.

72. Amendments were made to the standard FIDIC forms, particularly as regards the design and co-ordination responsibilities placed on a number of the civil contractors and the fact that the employer arranged blanket third party and builders risks insurance cover for the project as a whole.

73. The contract placed on the contractor the fundamental obligations and associated risks in completing the works in accordance with the employer's requirements. Contractors took the risk, *inter alia*, of changes in primary and secondary legislation, currency and price fluctuations, and costs associated with delays caused by inclement weather. The employer's risks included delays resulting from inclement weather, and costs and/or delays attributable to adverse physical conditions which could not have been foreseen at tender.

74. The contract provided the employer with a right to deduct liquidated damages against the contractor's failure to achieve sectional and final completion dates. Sectional completions (to a predefined degree of completeness) were neces-

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sary to enable the employer's trackwork contractor and system-wide electrical and mechanical contractors to begin work during the subsistence of the civil contract.

75. The law applicable to the contract was the law of the Republic of Singapore. Singapore's legal system is based on English Law, with Common Law contractual doctrines and remedies similar to those recognized in the UK.

Prequalification of contractors

76. Overseas and local contractors were advised to register their interest in prequalification by submitting details of their relevant experience, current technical and managerial expertise, financial standing, etc. Prospective tenderers were advised of the preferential margin schemes, thus encouraging liaisons between local and overseas contractors. As many of the early contracts (phase 1) were to be awarded on a 'design and construct' basis, contractors seeking prequalification for these contracts were required to demonstrate their design ability.

77. Approximately 400 contractors (singly or in joint venture) applied to be prequalified for phase I contracts. Between six and eight contractors were prequalified for each 'design and construct' contract, and between ten and twelve prequalified for each 'construct only' contract. As many contractors appeared to meet the necessary requirements, a prequalifying contractor was permitted to submit tenders for a maximum of only three contracts.

78. Similar prequalification exercises were carried out for phases 1A and 2.

Form of contract: design and construct

79. It was primarily with speed in mind that the early contracts were awarded on a 'design and construct' (or contractor's design) basis. This put the design and co-ordination responsibility on the contractor who, either by using in-house designers or by engaging consultant designers, was required to design the works in accordance with the employer's requirements (see below) and subject to design acceptance by the Engineer. While the design development process was underway, the construction work could begin and thereby shorten the overall time for construction.

80. Contractor's designs were based on general arrangement drawings, general and particular specifications and design criteria which had been prepared by the employer. Design development, co-ordination with numerous system-wide electrical and mechanical designers/contractors, and detailed architectural, civil and structural design including materials and workmanship specifications were the responsibility of the contractor. Design undertaken by the contractor was subject to the Engineer's acceptance as compliant with the employer's stated requirements. The contractor's design, co-ordination and project management costs were reflected in the lump sum tender price.

81. Apart from the programme advantages of the 'design and construct' approach, contractors were able to prepare their designs to suit preferred construction techniques and thereby to submit more competitive prices.

82. A further advantage was the resulting variety of architectural styles. All the stations between Bishan and Outram Park were designed by contractors, and each presents a unique architectural character.

83. The co-ordination responsibility placed on the civil contractors demanded the establishment by the contractor of a dedicated co-ordination team which was required to liaise with system-wide electrical and mechanical designer/contractors and thereby to ensure compatibility, efficiency and economy in design. Resulting

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from this co-ordination, civil contractors were required to produce combined services drawings (CSDs) and structural–electrical–mechanical drawings (SEMs), the latter showing the structural penetrations and openings necessary to accommodate the electrical and mechanical installations. When necessary, the Engineer was brought in to resolve co-ordination difficulties. The success of this system of co-ordination depended on the degree of importance which each main contractor placed on this obligation. On reflection, it is thought preferable for the Engineer to play a more active supervisory role and thus to encourage active co-ordination between the various contractors.

Form of contract: construct only

84. The Corporation was aware of the environmental impact of the overhead sections of the proposed railway. Early decisions emphasized this concern: for instance, the use of 'third rail' electrification in preference to an overhead power supply system. Aesthetic considerations as well as engineering factors led to the Corporation engaging its own design consultants to design the overhead stations and viaducts so as to unify the visual impact. Additionally, the speed of construction for works above ground is appreciably faster than works below ground. Time saved by overlapping design with construction was not significant in terms of construction completion and public opening dates. Consequently, all above-ground stations and viaduct works were awarded on a traditional 'construct only' (or Engineer's design) basis.

85. In addition, it was anticipated that local contractors would be well placed to undertake much of the works above ground, based on their experience in local elevated highway schemes. Local contractors would be encouraged to apply for construction prequalification without the added responsibility of design.

86. The contracts for the design of the main depot at Bishan and for the installation of trackwork were awarded as construct only contracts, the detailed design for the depot and the system trackwork having been prepared by specialist consultants engaged by the employer. Early studies indicated that the additional structural costs of providing for development above depots did not warrant serious consideration of such development.

Tender evaluation—design and construct contracts

87. Prequalified tenderers were invited to attend briefing meetings with the employer and his advising consultants, during which tender requirements were explained and contractor's queries clarified.

88. A period of approximately four months was allowed for the tender. During this period, the tenderer was required to familiarize himself with the extent of the works and the site conditions (contractors were encouraged to carry out additional site investigation and many did so), and to prepare an outline design.

89. Tenderers were required to submit a lump sum bid which was fixed and therefore subject to change only by the issue of Engineer's instructions to vary the works (Variation Orders). Payments were to be made against completed activities, each tenderer submitting his cost loaded activities programme.

90. Lump sum bids were converted to a net present value which took due account of cash flow and eliminated any advantage which might have been gained by a tenderer who submitted a low, albeit front-end loaded, tender sum.

91. Acceptance of design and construct tenders was deemed to be an acceptance of the contractor's design concept only. All designs submitted at tender were

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required to be submitted to the Engineer for detailed examination and formal acceptance.

92. Tenderers were required to submit their tenders in a number of separate packages.

Package 1. Conditions of contract, design criteria and specification
Package 2. Employer's drawings
Package 3. Form of tender and appendix; offers for adjustment of contract price
Package 4. Tabulation of cost by activity, by major items of work, by calendar month, unit prices or rates
Package 5. Tenderer's programme
Package 6. Details and qualifications of design, supervisory and management staff for the project; details of proposed sub-contractors
Packages 7 and 8. Details of tenderer's drawings, computations, specifications and procedures for activities which were to be performed between receipt of a Letter of Intent and the design acceptance of the first stage submission
Package 9. Details of civil engineering works
Packages 10 and 11. Details of permanent and temporary structural, electrical and mechanical works
Package 12. Details of proposed sub-letting; details of proposed export credit arrangements.

93. At tender, architectural details and materials and the extent of finishing works was uncertain. Provisional sums were included in the tenders with each tenderer selecting a percentage on-cost (i.e. management fee) to be added to sub-contract tender sums and paid to the main contractor.

94. With such an arrangement, the employer had a financial interest to ensure that the architectural sub-contracts were awarded on compatible conditions and on a competitive basis. The contract empowered the employer to reduce or increase the number of sub-contract tenderers and to scrutinize the sub-contract documentation, thereby ensuring that items of work and obligations for which the main contractor was being paid in his lump sum were not inadvertently passed on to an architectural sub-contractor; otherwise the employer would have paid twice for the same item of work. Furthermore, the employer had the opportunity of accepting or rejecting the architectural sub-contractor proposed by the main contractor. In several cases, the main contractor (or a constituent joint venture partner) wished to bid for the sub-contract works (which in some cases were sizeable contracts in themselves). The employer's agreement to accept the main contractor in the list of sub-contract tenderers was given on the understanding that financial assessment of the sub-contract tenders was to be carried out solely by the employer.

95. Sub-contract nomination was considered but not adopted in that the employer did not want the main contractor's fundamental responsibility for the execution of the works to become contaminated by the performance of nominated sub-contractors.

96. In cases where works, which in scope were part of the architectural provisional sum, needed to be consistent throughout the system, arrangements were devised whereby the employer advised those main contractors calling sub-contracts that a single sub-contractor (of the contractor's choice) would be

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required to enter into sub-contracts with each main contractor. In such cases, the tenderers were advised that it was 'all or nothing' so that the employer benefited from any price reduction for quantity. In instances where the main contractor had not been appointed, the employer contracted with the system-wide sub-contractor on the understanding that the contract would be transferred (a novation made) to the successful main contractor. Main contract provisions were drafted accordingly.

97. Evaluation of the tenders was scheduled to take approximately six weeks. In most cases, tenders were 'equalized and clarified' by a series of questions and answers, many of which were later incorporated into the contract agreement.

Tender evaluation—construct only contracts

98. Tenderers bidding for construct only contracts were allowed approximately two months to submit their tenders. These were priced against bills of quantities prepared by the employer. Quantities were variable and rates were fixed.

99. Tender evaluation was simpler than for the design and construct contracts in that each tenderer was pricing an identical document. Tender evaluation took approximately four weeks. As before, formal questions were put to tenderers for which written answers were required and, in most cases, incorporated into the contract agreement.

100. Tenderers were permitted to submit alternative designs and offer corresponding reductions in price. Relatively few of these alternatives were accepted.

101. Sub-contract arrangements were similar to those for the design and construct contracts.

Particular contract provisions

102. No provision for price fluctuations was incorporated into the contracts; this risk was to be borne by contractors. However, this was not considered to be too onerous as the construction industry recession was expected to continue, resulting in low construction inflation.

103. Tenderers were encouraged to submit financing offers with their tenders, although this was not mandatory (government resources were adequate and available). These offers were made in various forms, many tenderers engaging the services of merchant banks. Offers were made involving credit, deferred payments and payments in mixed currency (acceptable currencies were stipulated by the employer after consultation with the Monetary Authority of Singapore).

104. Contractors were paid a mobilization advance equal to 5% of the contract sum. The advance was recovered by the employer in equal instalments by deductions from the first six payment certificates.

105. The contract required the copyright in all designs to be the property of the employer. This will facilitate maintenance work and replacement, particularly for the electrical and mechanical elements of the project.

MRTC organization

106. Figure 11 shows the organizational arrangement of the MRT Corporation and, in particular, the Project Division. At the peak the Civil and Structural Department, embracing all architectural and engineering design and construction, supervision employed some 1100 staff.

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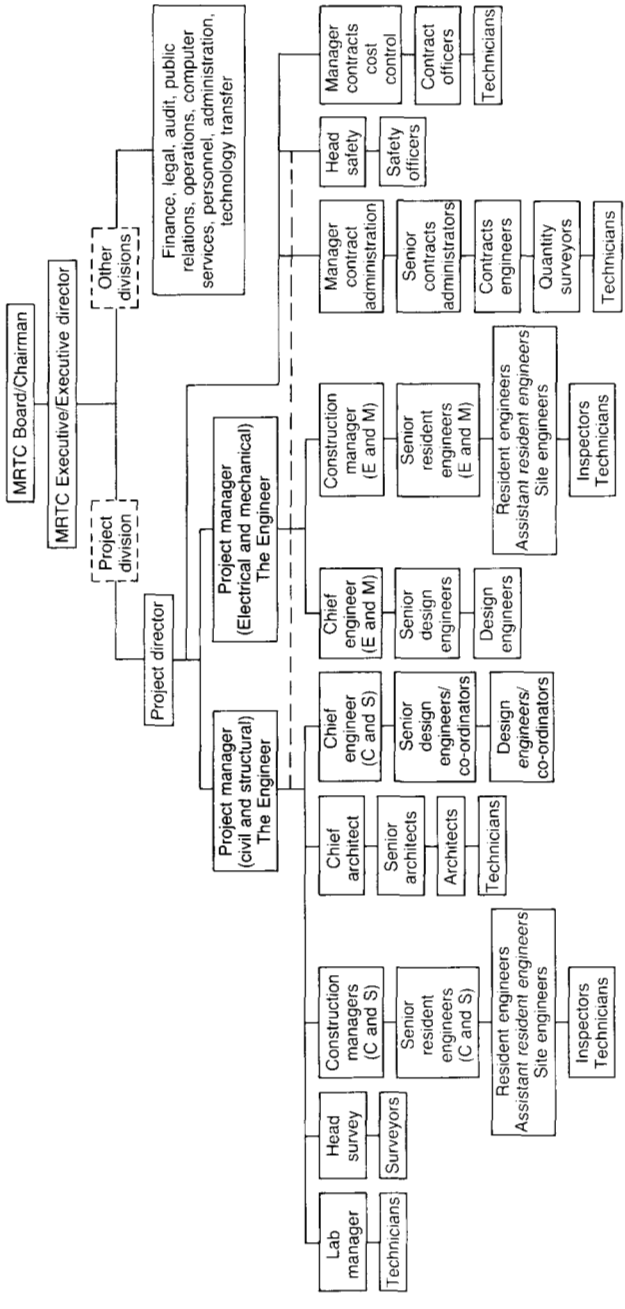


Fig. 11. MRTC organization chart

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107. For civil and structural works, a project manager was appointed and nominated as the Engineer, the latter's authority under the contract following the FIDIC arrangements. Six construction managers were engaged, each responsible for approximately five main civil contracts (a section). The construction costs for a section averaged about S$460 million (£141 million). Senior resident engineers, each being responsible for up to three contracts (a unit), were nominated as the Engineer's representative. The various and necessary powers of the Engineer were delegated to the construction managers and the senior resident engineers.

108. A Contracts Administration Department was established serving civil, structural, electrical and mechanical disciplines. For civil works, a staff of approximately 120 engineers and quantity surveyors was engaged. Teams were assigned to sections and based on site. A head office team monitored and co-ordinated effort and advised the Engineer in the discharge of his functions.

109. Design review and consultancy management departments were established under a chief civil engineer and a chief architect. All design was to be carried out by contractors or consultants and, therefore, the Design Department was responsible for review of submissions and for advising the Engineer on the acceptability of the proposed designs (non-acceptance requiring correction and resubmission from the contractor or consultant). The Engineer's powers relating to design were delegated to the chief civil engineer. The design and construct contracts were clear in that acceptance by the Engineer did not relieve the contractor of any of his contractual design obligations and responsibilities. For these contracts, contractors were required to submit their design of the permanent works in three distinct stages. The first stage, known as intermediate, required the contractor to submit a developed concept with supporting calculations. Sufficient detail was necessary to permit construction to begin should the submission prove acceptable. The pre-final submission required the contractor to submit those designs which were inadequate or incomplete at the earlier stage, and thereby to obtain the Engineer's in principle acceptance of the whole design Final design submission was essentially a resubmission of previously accepted designs with any minor amendments and corrections. The Engineer's formal acceptance was given when all aspects of the final design submission had been checked and found acceptable. Dates for submission of each stage were specified in the contract, as were the durations of the Engineer's review period. The interface between civil contracts, and between civil and electrical and mechanical contracts meant that the design submissions were hampered by the effects on civil contractors of late changes by system-wide electrical and mechanical contractors. After the Engineer's acceptance of the civil contractor's design, future changes (unless resulting from inadequate design by the civil contractor) were deemed variations for which the civil contractor was paid.

110. Architectural submissions were reviewed by a panel of eminent local architects who advised the Engineer. This was considered important as many designers were from overseas and the employer wanted to ensure that local architectural tradition and style were incorporated into the station designs.

111. Similarly, a panel of local artists and critics was set up by the Corporation to commission works of art (sculptures, murals, mosaics, mobiles) for eventual display in stations. Fourteen commissions were awarded to artists all based in or near Singapore.

112. A Materials Laboratory was established for testing samples from the works and for advising the Engineer on the suitability of proposed materials. A

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Safety Department was set up to ensure statutory regulations were followed and that each contractor fulfilled his contractual and public safety obligations. The safety record of the MRT project has, so far, been good compared with the local construction industry's record as a whole.

113. A survey section arranged mapping and survey control for the project. Survey teams were assigned to sections of the works to ensure the accuracy of setting out and tunnel shield control. Wriggle surveys to ensure track alignment could be maintained were carried out throughout the underground sections of the railway.

Contract administration and cost control

114. The Project Division of the MRT Corporation is responsible for administering and inspecting approximately 150 contracts. Since work on the project began in 1983, contractors have been paid an average of S$2 million (£0.6 million) per day. The contract administration and cost control efforts have needed to be substantial to ensure that the employer's obligations (many discharged by the Engineer in the capacity of employer's agent) are met and that costs are prevented from escalating.

115. The main contract administration tasks included the following

(*a*) ensuring adequate and relevant site records are kept
(*b*) ensuring Engineer's Instructions are issued in the correct form and in accordance with the Conditions of Contract; consequently, ensuring that any instructions constituting variations to the contract requirements are valued and that the contract sum is adjusted accordingly
(*c*) checking the contractor's applications for payment and advising the Engineer accordingly
(*d*) advising the Engineer on suitable rates for items of work omitted from the bills of quantities
(*e*) analysing of the contractor's claims for additional payments or extensions of time submitted under the provisions of the contract, and advising the Engineer accordingly on principle and quantification of compensation due
(*f*) analysing of contractual issues which have been submitted to the Engineer for his personal decision when an earlier decision by one of his delegates is challenged
(*g*) drafting correspondence on behalf of the Engineer or the employer in response to contractual issues raised by the contractor
(*h*) advising the employer during negotiations in which contractual differences are compromised by means of a supplemental agreement to the contract, or when mediation, conciliation or arbitration proceedings have begun
(*i*) preparing certificates of completion and maintenance
(*j*) advising the Engineer on matters concerning backcharges arising from the contractor's obligation to compensate the employer for any loss or damage
(*k*) preparing of final contract close-out reports and advising the employer that the contract has been fully executed by performance

Additional tasks related more to the internal procedures necessary in a large organization. The employer required regular financial reports, compliance with

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stringent limits of expenditure authority, annual financial and manpower budgets, compilation of fixed asset accounts, statistical returns to government departments, insurance reports, and so on.

116. For management of variations the employer chose to adopt a procedure whereby an originator, who may be a designer, design co-ordinator, construction supervisor, contract administrator or any one of the employer's many staff or agents, completes a request for variation sheet stating the proposal, its purpose and its estimated cost. A justification questionnaire is completed and submitted with the request. Each request for variation is numbered with its contract number and a serial number. After initial vetting for completeness and contractual accuracy, the request and its supporting papers are circulated to specified managers who have an opportunity of assessing the benefits, burdens and consequences of the proposed variation and the value for money. The responsibility for the final acceptance of the proposal lies with the project director, the Executive Committee of the MRTC or the MRTC Board, depending on the level of financial authority necessary. After acceptance, which signifies that the client wishes to proceed with the variation, the Engineer or his delegate issues the contractor with an instruction to vary the works.

117. In accordance with the contract, variations are valued using existing rates in the contract schedules whenever applicable, otherwise new rates have to be devised. If possible, the price of a valuation is agreed with the contractor before the variation order is issued. In cases where agreement could not be reached, the Engineer would value the variation order, and the contractor would, on completion of the varied work, be paid that amount.

118. In the five years 1984–88 there have been nearly 6000 instructions to vary the works issued on the MRT project; this approximates to over four variations for each working day since the first MRT construction contract was awarded. The average value of a variation is S$32 000 (£9600).

119. The immense volume of interrelated variations has required a computerized variations management system, devised and operated by the Contracts Administration Department, which can pinpoint the location of any proposed variation request and indicate its status. It is therefore possible to spot 'log-jams' and thus to expedite the process of approval.

120. In common with most standard forms, the MRT contracts require the contractor to submit payment applications. The Engineer, after review of the applications and their supporting documentation, determines the amount due and enters this amount in the Engineer's payment certificate. The employer will pay this amount subject to any financing agreements or set-offs. The contract provided the Engineer seven days to certify and the employer a further 21 days in which to make payment, following receipt of the application.

121. For the 'design and construct' contracts, payment applications were submitted on completion of specified, costed activities. No provision was made for payments related to measured works (except for some of the architectural subcontracts). On the 'construct only' contracts, monthly payments were made against interim measurements. It is impractical to expect a full measurement of all items each month (especially when the contract calls for the engineer's certification within seven days of receipt of the payment application) and agreed proportional estimates are often used as a basis of payment, with final measurement undertaken on completion of the work and any adjustments made at that stage.

122. To date, the employer has received about 1600 formal claims from the

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contractors under the provisions of the contract. Four-fifths of these are from civil contractors. The total amount claimed by contractors is in excess of $300 million (£90 million), and a likely 'payout' is about $100 million (£30 million). An elaborate procedure for claims handling has been devised: this begins with the numbering of the claim on receipt of notification and ends with either a final binding decision from the Engineer (if not challenged within the stated time period) or arbitrator, or a commercial settlement.

123. The contracts provide for the appointment of an arbitrator with extensive jurisdiction. There are currently several arbitrations in process involving MRT main contracts. It is interesting to note that the claimant in some of these actions is the employer who seeks to overturn the Engineer's decision given in favour of the contractor. It is open for the parties to negotiate and compromise quite independently from the claim resolution provisions of the contract. Negotiations may benefit from the services of a Mediator or Conciliator and there has been one mediation on a major tunnelling claim.

124. To enable contracts to be closed, a checklist of approximately 65 items, which need to be concluded, has been drawn up. Agreement and payment of the final account is an essential part of the close-out procedure. The close-out report consists of a record of performance of the various obligations resting on the contractor, the employer and the Engineer.

125. A *Procedures Manual* was first published in early 1984 and consists of numerous flow charts, specimen letters, sample forms, etc. The Manual, which has expanded during the project and is now published in five volumes, aims to unify and regulate contract administration effort, and thereby to simplify management's grasp of the often complex issues associated with the contracts.

126. Extensive use has been made of computers in the contract administration and cost control of the project. Access to computer data has been made available to various staff at headquarters and on site, security of information being maintained by issue of personal passwords. Systems employed include: engineers instruction; variation requests and variation orders; payments; claims under the provisions; insurance claims; contract financial abstract and most likely final costs.

127. Under the control of the Contracts Administration Department, a Technical Library was established in 1984, consisting of 1000 specialist volumes and 1250 standards, and subscribing to 89 technical periodicals. Interlibrary loan facilities with the other main technical libraries in Singapore have been arranged. Also, a Records Centre was established in 1987, when a programme of microfilming and computer indexing of project documents and drawings was started.

128. While many of the contract administration procedures and policies which have been adopted for use on the MRT project cannot be said to be the only satisfactory approach, it is predicted that the final cost will exceed the tender cost by only approximately 8%; for a fast-tracked, technologically unique MRT system built on or through very mixed and unstable ground, this is a reflection of the adopted procedures, close co-operation and concerted efforts of all those individuals and organizations who have been involved in the project.

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